SENSEYE PREDICTIVE MAINTENANCE

Predictive Maintenance is about more than algorithms

How Senseye Predictive Maintenance focuses on meaningful outcomes for users.

Find out more: **siemens.com/senseye-predictive-maintenance**

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Introduction

Approaches to predictive maintenance that are driven by machine learning rely on the automated analysis of condition monitoring data to track an industrial machine's behavior over time. This allows algorithms to make increasingly accurate forecasts about what is likely to happen next. Combining this machine-centered model with a model that also learns about the priorities of maintenance teams helps to optimize the decision-making support that predictive maintenance solutions provide – which should ultimately be the focus.

Most experienced maintenance engineers already have a detailed mental picture of the machinery they care for. They know, for example, whether a rattling valve means a breakdown is imminent, or if it is safe to ignore it until the next scheduled shutdown. If a predictive maintenance system can tap into this pool of knowledge, it can learn to provide the best possible support in the decision-making process.

In a user-centric approach to predictive maintenance, all the information about evolving machine behavior is still safely processed and stored in the system in case it is needed. However, maintenance teams are alerted only when the system thinks they will find that information useful, based on what they have found helpful in the past. This is the core of how Senseye Predictive Maintenance works, ensuring that users are not bombarded with information that they find of little interest and value.

Get set to succeed in a changing world

Every predictive maintenance initiative – from an iron rod held to a rattling machine to a sophisticated automated software monitoring system – does the same job of conveying information to help operators identify when trouble is brewing. The chief aim is to flag up issues early enough to prevent a breakdown that would otherwise lead to costly unplanned downtime. There are often other benefits too, related to improved productivity and maintenance planning.

It is impractical in most industries to build a detailed and robust mathematical model specific to each individual machine, so most predictive maintenance systems work instead with limited data inputs from the factory or plant floor evaluated against simple alarm levels. They might alert users when something is heating up or vibrating more, but they are unlikely to have enough information to make a detailed diagnosis. Manually setting alert thresholds and alarms using these inputs can be of great benefit at smaller scale. However, it is another process to maintain and quickly becomes unwieldy with more than a handful of assets.

This makes the challenge for the user of deciding whether they should respond immediately to an alert or whether it is safe to leave it for another day all the more difficult.

In other words, the system can advise, but only the user has the experience and expert knowledge to decide when to act – provided they are not overwhelmed with information resulting in their attention becoming misplaced.

Senseye Predictive Maintenance is a low-context application designed to provide decision support to this type of knowledgeable user, without the need to manually build a dedicated 'digital twin' of each asset. It accepts a range of inputs – from dedicated condition monitoring data to more general plant data – and looks for patterns or characteristic signals that indicate that there may be a problem. Its failure-based forecasting capabilities work especially well when failures leave distinct signals or 'fingerprints' in the data, as well as where the data is characterized by regular repeating patterns. In many cases and with sufficient data, Senseye Predictive Maintenance is able to calculate the remaining useful life (RUL) of each asset – a technique known as prognostics.

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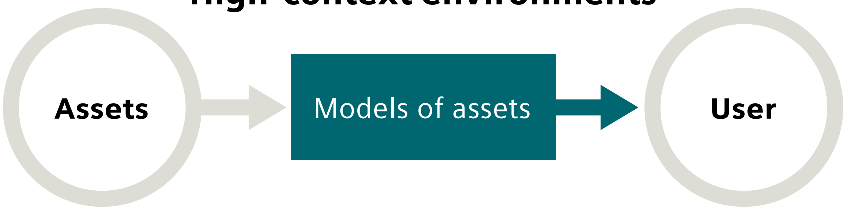
High-context environments



Highly contextual environments (e.g. where schematics and excellent quality data are available) lend themselves well to focusing on the asset in detail.

Low-context environments

Low context environments, where the asset data is 'simple' process data or the asset itself is inexpensive (although the cost of failure and maintenance may not be) lend themselves well to modelling the user and what the user would find interesting and valuable.

Senseye Predictive Maintenance is designed to make life easier for customers' data scientists and machine experts, not replace them. It monitors assets at such a scale where it would not make economic sense to build and maintain custom models. However, it is designed to integrate with and allow refinement by custom models that users already have or wish to develop to further enhance the outputs.

Lessons from life

Digital content providers such as Netflix or Amazon routinely store information about what each user chooses to watch. They then use that information to drive a 'recommendation engine' that refines the movies it shows to each user continuously in response to their feedback.

Outcome-oriented predictive maintenance works in a similar way, although it is responding to feedback from all the operators to decide upon what is useful and appropriate to maintainers and engineers.

From data to **user priorities**

Senseye Predictive Maintenance has introduced Attention Index in the Senseye Predictive Maintenance cloud-based predictive maintenance solution as a way of shifting focus onto what matters to maintenance engineers.

Whenever Senseye Predictive Maintenance raises an alert, the operator can indicate at the touch of button whether that alert is useful or not. Over time, this teaches the system to direct the operator's attention towards the most pressing maintenance priorities, rather than bombarding them with low level alerts from all directions. This is especially helpful in major deployments that cover hundreds or even thousands of assets.

The system is learning all the time. The number of alerts will be higher in the first few weeks of a Senseye Predictive Maintenance deployment, but as users feed-back information about which alerts are helpful, it will gradually stabilize at a level that directs attention only where it is needed – without requiring the user to have training or knowledge of data science techniques.

Senseye Predictive Maintenance is unique, user-centric approach to data analytics is based on three principles: guide attention, focus on meaning and model the user.

Guide attention

A series of pattern engines identify chunks of data with meaning, feeding the results back into the attention engine, which predicts how interesting these patterns are likely to be to the user. The result is Attention Index.

Each pattern engine is a collection of algorithms that are designed to work well together 'out of the box' but can also be configured by the user to improve their accuracy even faster.

The attention engine estimates Attention Index for each asset by analyzing recent patterns, taking into account previous patterns, feedback, asset criticality, configuration settings and the production schedule.

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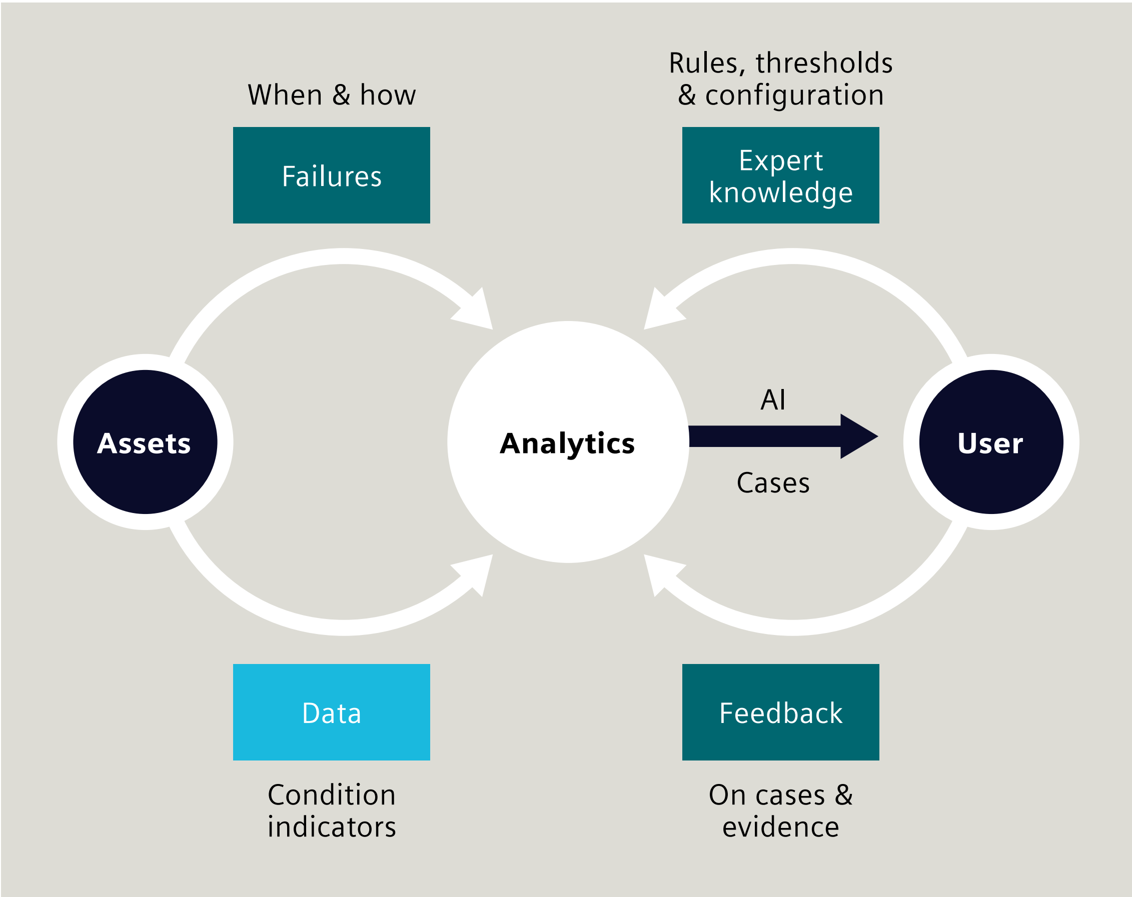
Focus on meaning

Good data is essential for optimal analytics, as are high-quality algorithms. Context is also extremely useful but can be scarce in real-world environments. Senseye Predictive Maintenance therefore focuses on designing analytics that can work with the minimum amount of context, using limited context as efficiently as possible and tapping into user expertise to glean as much context from them as possible.

Model the user

Focusing the analytics around the user makes the best possible use of scarce context. Instead of predicting 'machine health', which is a property of the asset, Senseye Predictive Maintenance predicts the level of user interest, which is a property of the user. At each interaction with the system, feedback enables the application to build an increasingly accurate model of the 'average' user in each deployment.



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Practical support for real-world environments

Senseye Predictive Maintenance is designed to be able to work without any upfront rules if necessary, but users can also opt to prime it with a range of information, such as thresholds, important trends or data from past failures. It can even take other factors such as criticality into account in addition to the condition of the machinery. For example, if one welding robot with a hotbackup starts to malfunction it will not have much impact on overall productivity. In contrast, if the sole forging machine in a factory stops working it could result in a major production bottleneck.

Such considerations and the expert knowledge of experienced operators are used in Attention Index calculations from the start.

Attention Index provides a single way of sorting assets that takes account of all the ways that Senseye Predictive Maintenance can detect or predict issues. If Attention Index reaches the level of a warning, the display also makes it clear why. In other words, Senseye Predictive Maintenance is already guiding the user towards the underlying problem before they have even looked at the case in detail.

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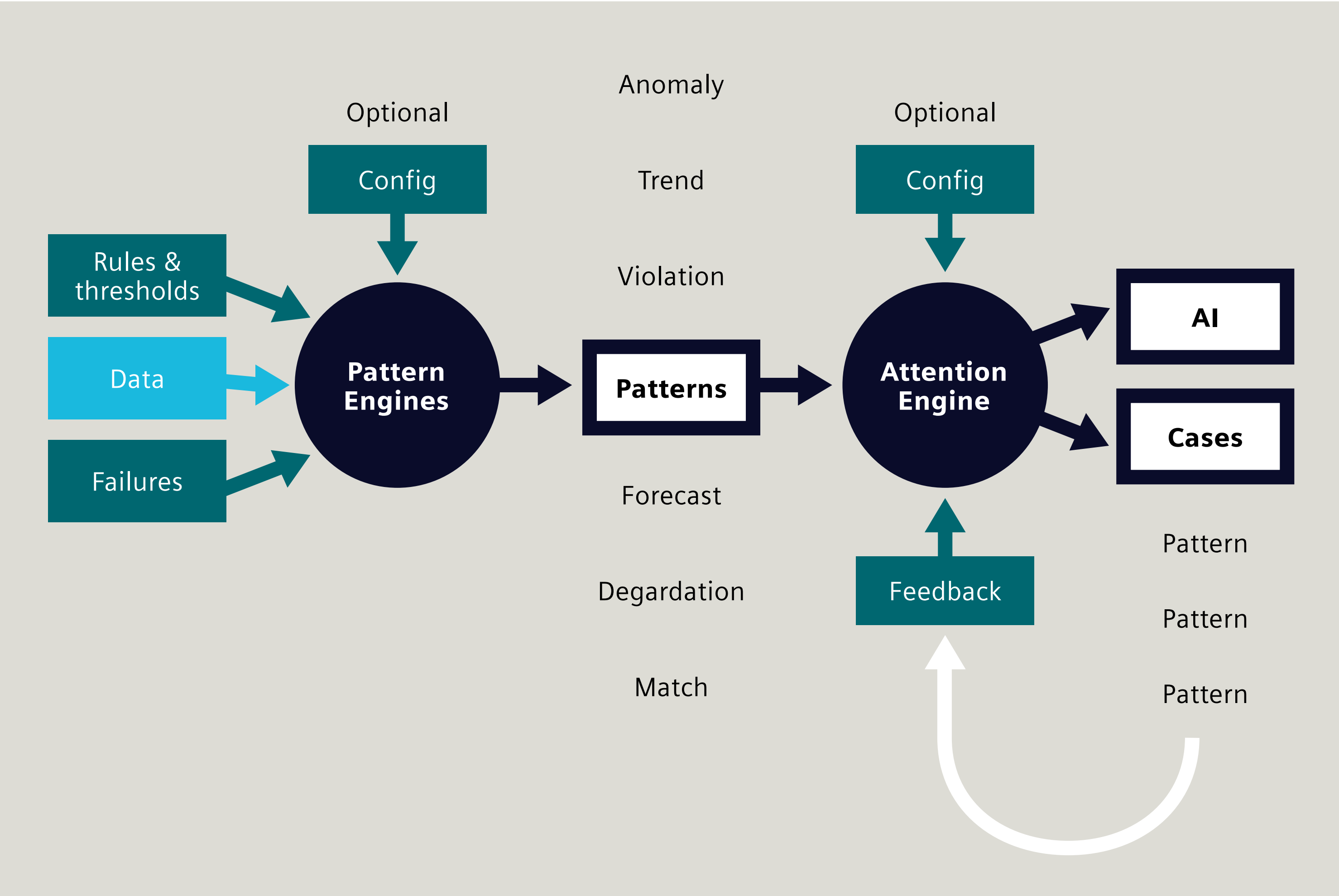
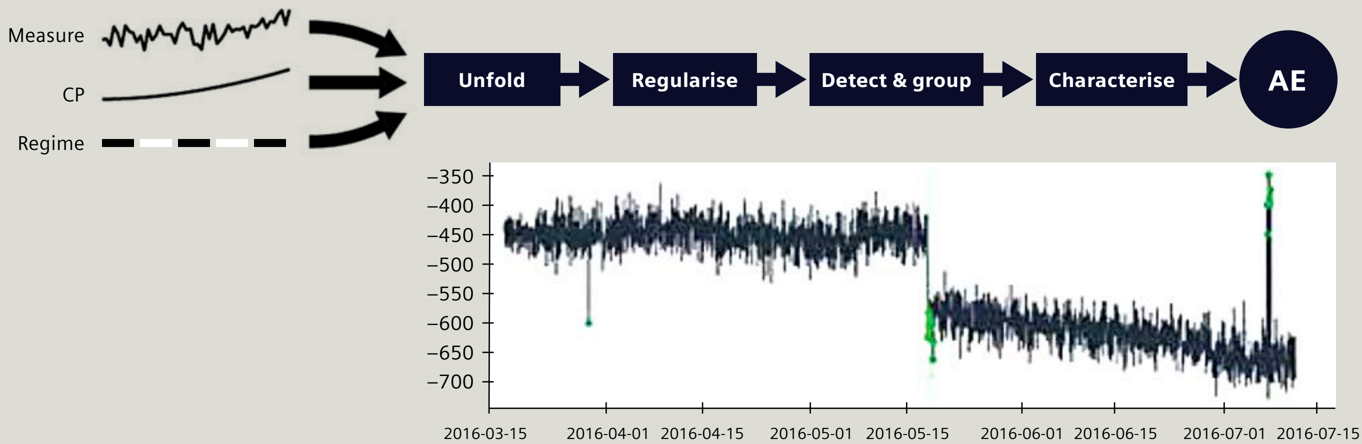
Spotting tell-tale patterns in data



Data analytics is about searching for patterns within the mass of incoming plant data. In the case of the pattern engines behind an Attention Index alert, those patterns fall into several distinct categories, which each feed results back to an overall attention engine. It's important again to highlight here that all of these methods and techniques are automated, completely transparent to the user so that they don't need to worry about the underlying mechanics of Senseye Predictive Maintenance as well as the machines under their care.



Anomalies are periods of unstable data. The anomaly pattern engine looks for anomalies in one or more parameters.
It comprises three components:

- Regularization deploys neural networks and various statistical techniques to remove dependencies between measures, as well as other artefacts that might otherwise interfere with detection.
- Detection builds a model of normal behavior using an adaptive window and uses statistical techniques to compare the most recent data point to the data in the window.
- Merging uses heuristics to filter anomalies and merge them into anomaly patterns that are characterized and passed up to the attention engine.

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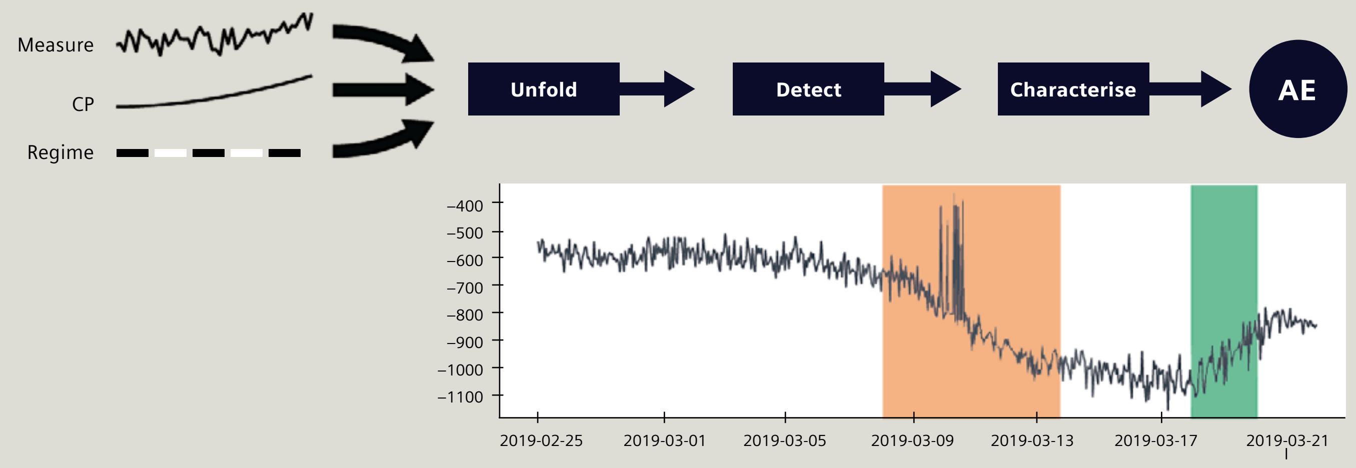
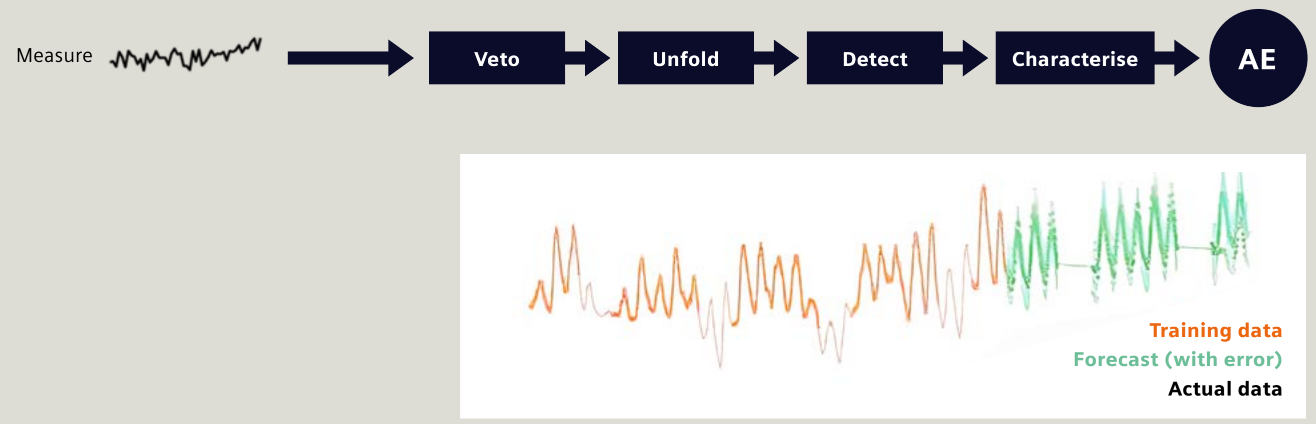


Trends are gradual shifts in the baseline. The trend pattern engine looks for gradual changes in one or more parameters.

- Regularization removes dependencies between measures, as well as other artefacts.
- Detection detects and extends trends using statistical techniques that account for the 'unusualness' of any candidate trend.



Violations of thresholds or rules can be specified by the user, so the violation pattern engine can spot them.

Senseye Predictive Maintenance: **a risk-free solution**

Senseye Predictive Maintenance is a cloud-based, software-as-a-service solution, so it is easy to scale up or down to match customer requirements. It is also backed by Senseye Predictive Maintenance's guarantee: If deploying Senseye Predictive Maintenance fails to reduce unplanned downtime as agreed upfront, customers can claim a refund on their entire subscription fee.

Existing users include blue chip companies in manufacturing, heavy industry, automotive and FMCG, who typically enjoy a 50% reduction in unplanned downtime.

By helping users to identify where they should be focusing their maintenance resources, Senseye Predictive Maintenance enables true predictive maintenance and delivers major productivity gains:

- 50% reduction in downtime
- 55% increased productivity
- 85% increase in maintenance accuracy

Want to find out more about how Senseye Predictive Maintenance can help direct your maintenance efforts to where they'll do the most good? Get in touch today to see a demo and get started!

Find out more:
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